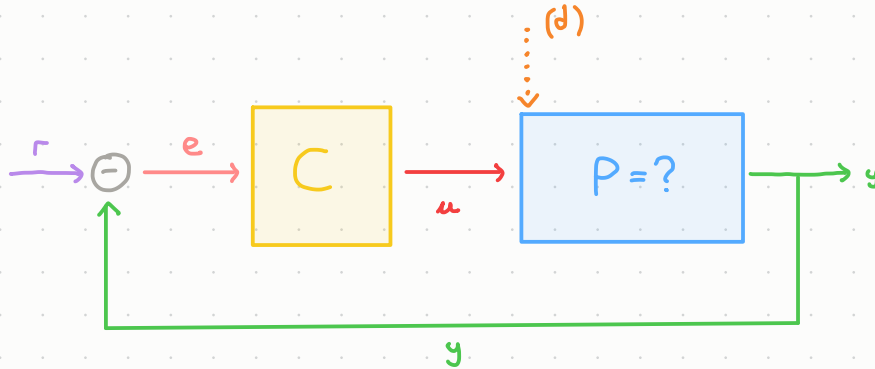


Exercise 2



Goal Today: Modeling, Block Diagrams

Last week: we built this system around P .



- We designed everything around it, but never looked inside...
- Today: what is inside $P \rightarrow$ a model.

② Warm up:

- You have Fresh coffee at 80°C in room temperature (20°C). How will $T(t)$ look like?



- You just predicted the future without any equations. Your brain has a model:

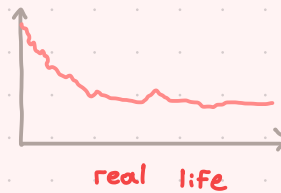
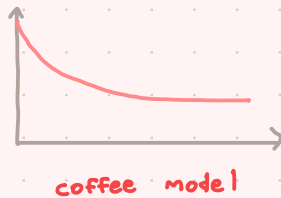
big T difference $\rightarrow$ fast cooling
small T difference $\rightarrow$ slow cooling

- Your intuition-derived model in math:

$$C \cdot \frac{dT}{dt} = -k \cdot (T - T_{\text{room}})$$

- As you just derived, a model is a simple set of rules/equations that describe how a system evolves.

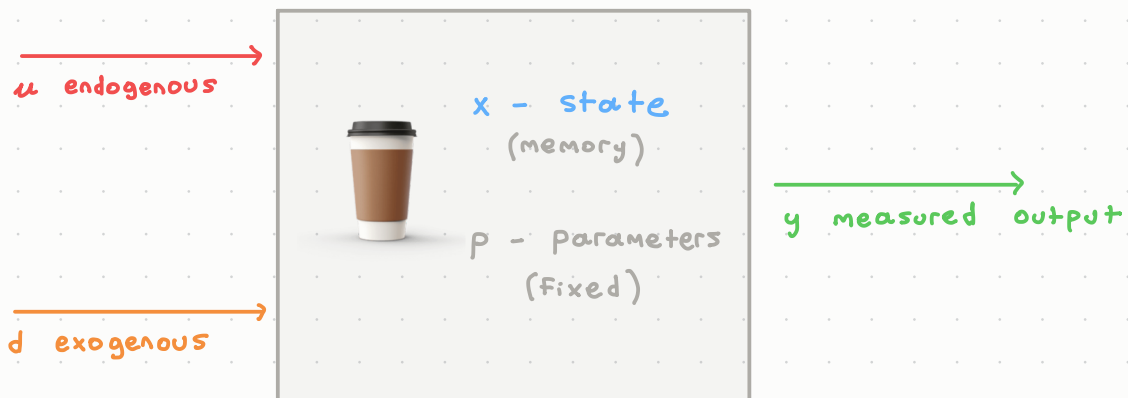
⚠ All models are wrong but some are useful:



Our Goal: Model a system

- ① draw the boundary of our system
- ② physics
- ③ write it in standard form

① Boundaries - who is who?



- u - inputs I can manipulate: endo. inputs
- d - inputs the world manipulates: exo. inputs / disturbances
- x - what the system remembers
- p - what never changes

○ For our coffee example:

- x - what is stored? $\left\{ \begin{array}{l} T \text{ of coffee} \\ T \text{ of mug} \\ m \text{ of coffee} \\ V \text{ of coffee} \end{array} \right\}$ are all possible.
- u - what can I do? $\left\{ \begin{array}{l} \text{pouring in cold milk} \\ \text{inputting/taking away heat} \\ \text{adjusting room } T \end{array} \right\}$ are all possible.
- d - what plays against me? $\left\{ \begin{array}{l} \text{wind} \\ \text{room } T \\ \text{sun shining} \end{array} \right\}$ are all possible.

- Notice how there are multiple possible modeling options. Choosing the right one depends on what we care about and/or what makes modeling easier.

if I care about how much coffee is left:

$$x(t) = m_{\text{coffee}}$$

if I care about the T of our coffee:

$$x(t) = T_{\text{coffee}}$$

or I care about both:

$$x(t) = \begin{pmatrix} m_{\text{coffee}} \\ T_{\text{coffee}} \end{pmatrix}$$

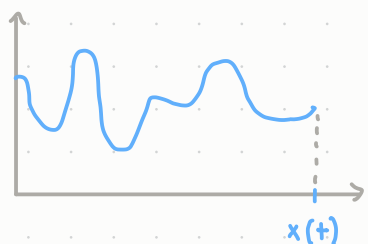
but we wouldn't use something like:

$$x(t) = \begin{pmatrix} m_{\text{coffee}} \\ V_{\text{coffee}} \end{pmatrix} \rightarrow \text{unnecessary}$$

- If our system takes in multiple inputs $u_1, u_2, \dots$ or has multiple outputs $y_1, y_2, \dots$ we call it a MIMO system. If only one u and one y , then it's SISO. In this course we only deal with SISO systems.



- The state is the smallest piece of information that summarizes the entire past:



We poured 300ml coffee, drank 100ml, spilled 20ml, added 200ml milk, and now we have 200ml of coffee

} past
} state x

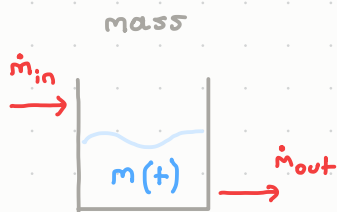
? Your turn: electric Kettle

- Water temp
- Amount of water (assuming closed system)
- Power on/off
- Heat loss
- Stable room temp.
- Thermostat reading

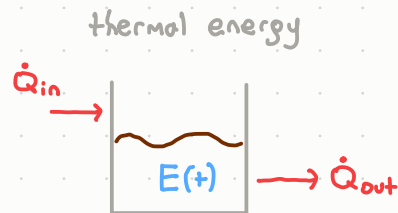
② Physics (where do our equations come from?)

- The bathtub law is an intuitive way to model systems:

$$\frac{d}{dt}(\text{storage}) = \sum \text{inflows} - \sum \text{outflows}$$

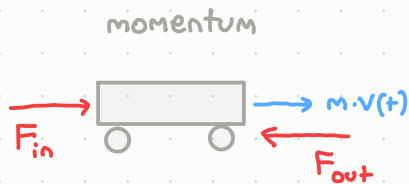


$$\frac{d}{dt} m(t) = \dot{m}_{in} - \dot{m}_{out}$$



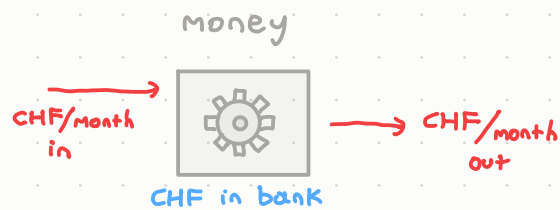
$$\frac{d}{dt} E(t) = \dot{Q}_{in} - \dot{Q}_{out}$$

$$m \cdot c \cdot \frac{d}{dt} T(t) = \dot{Q}_{in} - \dot{Q}_{out} \Rightarrow \text{coffee example}$$



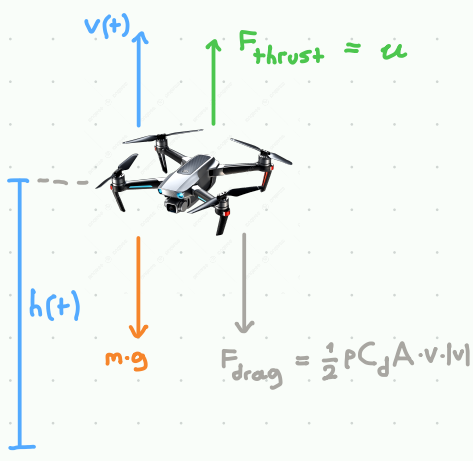
$$\frac{d}{dt} m \cdot v(t) = F_{in} - F_{out}$$

$$m \cdot a = F, \text{ you know this}$$



$$\frac{d}{dt} \text{CHF in bank} = \text{CHF}/\text{month in} - \text{CHF}/\text{month out}$$

③ Your turn: drone from last week



- let's model the drone for the state/storage quantities

$$x(t) = \begin{pmatrix} v(t) \\ h(t) \end{pmatrix}$$

③ Standard form / State-Space Form

- Now we have a **model** as some set of equations. Now we need to write our model in the "standard way".

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)) \quad \mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{u}(t))$$

↳ only first derivatives of $\mathbf{x}$

- So we need every equation in our model to be a first order ODE. But what if our model is governed by a higher order ODE, such as:

$$x'''(t) + a \cdot x''(t) + b \cdot x'(t) + c \cdot x(t) = 0$$

Remember LinAlg II:

- We can transform a high-order ODE into many 1st order ODE's:

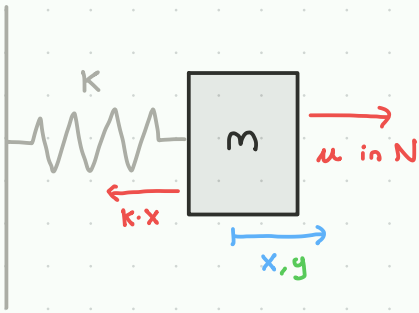
$$\begin{array}{lcl} x_0 = x & \Rightarrow & \dot{x}_0 = x_1 \\ x_1 = \dot{x} & & \dot{x}_1 = x_2 \\ x_2 = \ddot{x} & & \dot{x}_2 = -c \cdot x_0 - b \cdot x_1 - a \cdot x_2 \end{array}$$

- We describe the exact same system, but now we fulfill the standard notation.
- We actually did this with our **drone** without realizing:

$$\begin{aligned} m \cdot a &= F \\ \Rightarrow m \cdot \ddot{h} &= \underbrace{F_{\text{thrust}}}_{u} - F_{\text{drag}} - m \cdot g \\ \Rightarrow \ddot{h} &= u/m - F_d/m - g \quad (\text{actually a 2nd order ODE}) \end{aligned}$$

$$\Rightarrow \begin{array}{l} x_1 = h \\ x_2 = \dot{h} = v \end{array} \Rightarrow \underbrace{\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix}}_{\dot{\mathbf{x}}} = \underbrace{\begin{pmatrix} \dot{h} \\ \dot{v} \end{pmatrix}}_{\mathbf{f}(\mathbf{u}, \mathbf{x})} = \begin{pmatrix} x_2 \\ u/m - F_d/m - g \end{pmatrix}$$

? Your turn



$$m \cdot \ddot{x} =$$

$$\ddot{x} =$$

$$x_0 =$$

$$x_1 =$$

$$\dot{x}_0 =$$

$$\dot{x}_1 =$$

$$y =$$

- Now, if our model is **linear**, we can simplify it even further using matrices. Doing so results in the **state-space form**.

$$\begin{aligned} \dot{x} &= A \cdot x + B \cdot u \\ y &= C \cdot x + D \cdot u \end{aligned}$$

- We can do this on our spring-model.

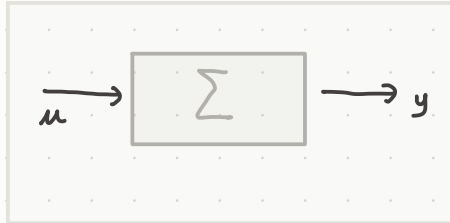
$$\begin{pmatrix} \dot{x}_0 \\ \dot{x}_1 \end{pmatrix} = \begin{pmatrix} \quad \\ \quad \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} + \begin{pmatrix} \quad \\ \quad \end{pmatrix} \cdot u$$

$$y = \begin{pmatrix} \quad \end{pmatrix} \cdot \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} + \quad \cdot u$$

- We can't do this on our drone, as it is **non-linear**. But we **can linearize non-linear** models! More next week...

Block Algebra

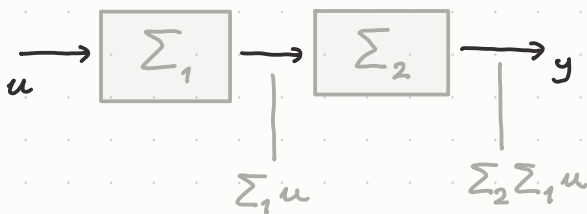
- In CS, we use blocks to symbolize plants and controllers. Thus, it is wise to know how to work with them.



- A block symbolizes a system which maps an input u to an output y . This mapping is shown as:

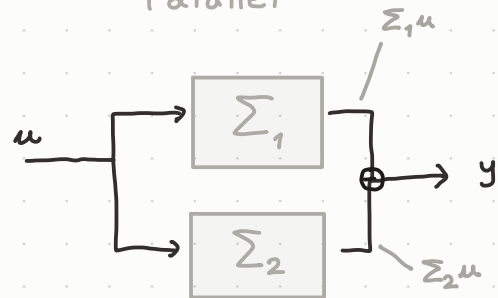
$$y = \Sigma u$$

Series, used in Feedforward Control



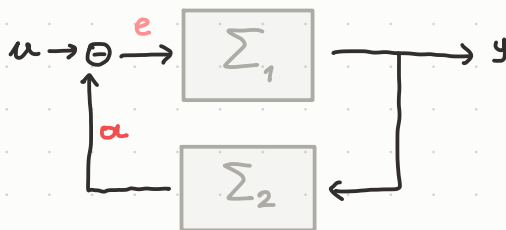
$$y = \Sigma_2 \Sigma_1 u$$

Parallel



$$y = (\Sigma_1 + \Sigma_2) u$$

Feedback



$$\alpha =$$

$$e =$$

- In general $\Sigma_1 \Sigma_2 \neq \Sigma_2 \Sigma_1$ because order generally matters. In CSI however, $\Sigma_1 \Sigma_2 = \Sigma_2 \Sigma_1$ because they are scalars.

? Exam Question

Problem: Consider the interconnected system shown in Figure 4. You can assume that the input-output relation for each system Σ_i is given by $y_i = \Sigma_i \cdot u_i$, that is, each of the systems Σ_i represents a simple scalar gain.

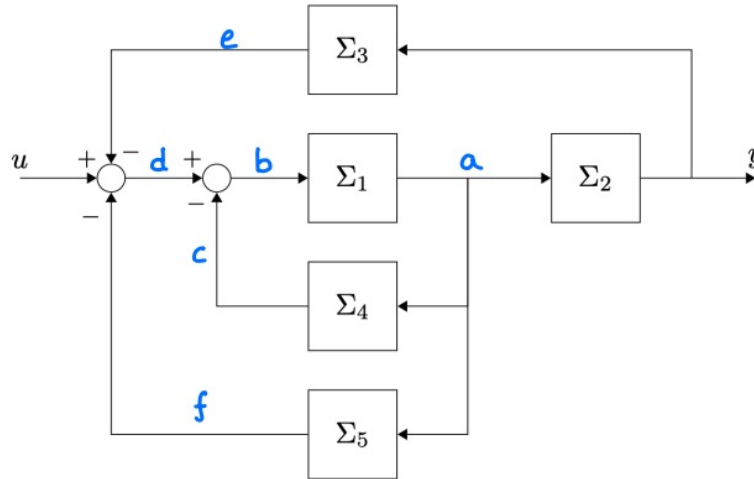


Figure 4: Interconnected system.

Q4 (1.5 Points) Derive the transfer function Σ from u to y for the interconnected system shown in Figure 4. **Simplify the result as much as possible.**

? Exam Question

Problem: Consider the interconnected system shown in Figure 1. The input-output relation for each system Σ_i is given by $y_i = \Sigma_i u_i$, that is, each of the systems Σ_i represents a simple scalar gain.

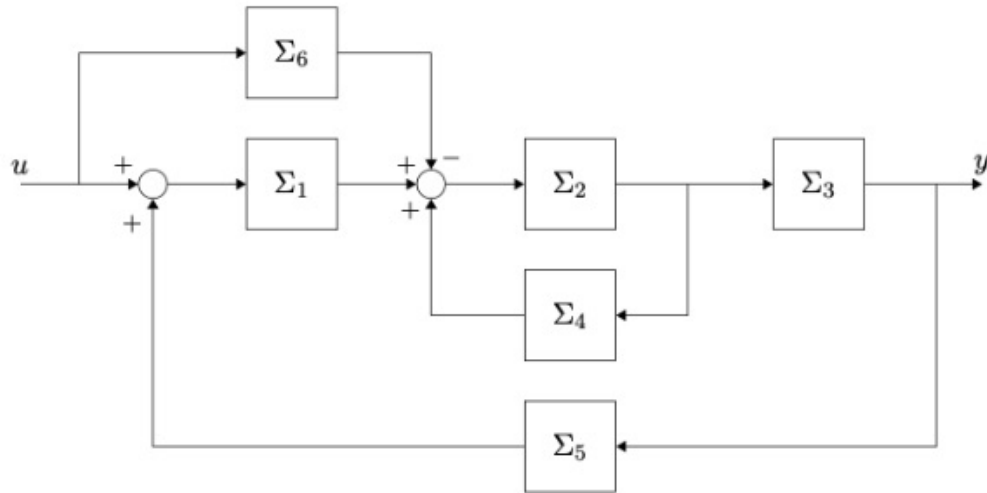


Figure 1: Interconnected system.